

Problem 2: Let V be a vector space containing vectors v_1, v_2 . Let $w_1 = v_1 - v_2$, $w_2 = v_1 + v_2$. Show that v_1, v_2 are linearly independent if and only if w_1, w_2 are linearly independent.

$\Rightarrow$ If v_1, v_2 are lin. indep., let $c_1 w_1 + c_2 w_2 = 0$. Then,

$$(c_1 + c_2)v_1 + (c_1 - c_2)v_2 = 0 \text{ implies}$$

$$c_1 + c_2 = 0$$

$$c_1 - c_2 = 0$$

$$\text{So } c_1 = c_2 = 0.$$

$\Leftarrow$) Similarly, if w_1, w_2 are linearly indep., assuming $c_1 v_1 + c_2 v_2 = 0$,

$$\begin{aligned} 0 &= c_1 v_1 + c_2 v_2 = c_1 \left(\frac{1}{2}(w_1 + w_2) \right) + c_2 \left(\frac{1}{2} \right) (w_2 - w_1) \\ &= \frac{1}{2}(c_1 - c_2)w_1 + \frac{1}{2}(c_1 + c_2)w_2, \text{ so that} \end{aligned}$$

$$c_1 = c_2 = 0.$$

Solutions

Quiz 1

You will have 20 minutes for this quiz, which consists of two problems based on lecture note pages 6-23. If you have any questions or concerns, please ask as soon as they arise. Good luck!

I hereby certify that this is my own work in accordance with the honor code and I have not collaborated with others or used any digital or analog resource during this assessment.

Name: _

SID: _

Problem 1:

Is the set

$$\{(a, b, c) \in \mathbb{R}^3 \mid a^2 = b^2\}$$

a subspace of $\mathbb{R}^3$? Justify your answer by either proving it a subspace OR showing that it fails a vector space axiom.

→ This is not a subspace because it fails to be closed under addition. The vectors $(1, 1, 0)$ and $(1, -1, 0)$ are in the space, but $(1, 1, 0) + (1, -1, 0) = (2, 0, 0)$ is not.